ENGINEERING PORTFOLIO

HANDS-ON PROJECTS & TECHNICAL DEPTH



Ajeeth Venkat Prasat

Embedded Systems Engineer

Biomedical wearables · IoT edge devices · Signal processing · Edge AI



TENG INSOLE OUTPUT · HEEL STRIKE / TOE-OFF · ≈ ±200 mV

01 SENSE
Transducers
TENG & IMU

→

02 CONDITION
Analog front-ends
High-impedance buffering, biasing, PCB design

→

03 COMPUTE
Firmware & systems
ESP32, STM32, Arduino, ROS, C/C++

→

04 LEARN
Edge AI
XGBoost, TinyML, FFT features

→

05 ACT
Insight
Clinical gait data, alerts, dashboards

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00 Executive summary

I build embedded systems that sit close to the physical world: sensors that produce awkward signals, the electronics that make those signals usable and the firmware + machine learning that turn them into decisions. Over three years across industry (Cognizant) and applied research (Fraunhofer EMI, IivMatS), I have taken projects from a blank schematic to a working, measured prototype.

This portfolio documents that work in more depth than a CV allows: the architecture of each system, the design decisions behind it, the results it achieved and what I would do next. My M.Sc. thesis at the University of Freiburg, a self-powered gait-analysis insole, is the flagship project and is covered first.

3 yrs Industry + applied research experience	**93.6%** Gait-activity accuracy, TENG insole + XGBoost	**2** Publications (IEEE Sensors J. under review, IRJET 2020)	**4** Working languages: English (C1), German (B1), Hindi, Tamil

CORE COMPETENCIES

EMBEDDED
Firmware & RTOS
C/C++ on ARM Cortex-M (STM32, ESP32), FreeRTOS, Linux, ROS

HARDWARE
Analog & PCB
Signal conditioning, power electronics, OrCAD & Altium layout

EDGE AI
ML on signals
FFT features, XGBoost, TinyML, TensorFlow, PyTorch

MEDTECH
Biomedical devices
Patient-safe low-voltage design, EMC, PoC prototyping

CONNECTED
IoT & protocols
CAN, SPI, UART, I²C, BLE, Wi-Fi, MQTT, LoRaWAN, cloud dashboards

CONTENTS

01 TENG gait-analysis wearable — 03
M.Sc. thesis · IivMatS / IMTEK, University of Freiburg

02 Autonomous sensor-fusion rover — 07
Student research assistant · Fraunhofer EMI

03 Smart waste monitoring with TinyML — 08
Start-up project · Hahn-Schickard / Uni Freiburg

04 Wireless IoT energy management — 08
Fellowship · Forge Innovation & Ventures

05 IoT sewage monitoring system — 09
B.E. thesis · Kumaraguru College of Technology

06 Professional experience — 10
Cognizant · Fraunhofer EMI · IivMatS · Forge

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CAREER PATH

NOV 2024 – NOV 2025
Master's thesis
IivMatS, University of Freiburg

FEB – JUL 2024
Student research assistant
Fraunhofer EMI, Freiburg

AUG 2023 – JAN 2024
Start-up project, TinyML
Hahn-Schickard, Freiburg

OCT 2022 – MAR 2026
M.Sc. Embedded Systems Eng.
University of Freiburg

MAR 2021 – AUG 2022
Software engineer
Cognizant, Bengaluru

JUL – DEC 2019
Graduate innovation fellow
Forge Innovation & Ventures

AUG 2017 – JUN 2021
B.E. Electrical & Electronics Eng.
Anna University (Kumaraguru College of Technology)

PROJECT 01 MASTER'S THESIS · LIVMATS & IMTEK, UNIVERSITY OF FREIBURG · MAR – OCT 2025

An intelligent, energy-aware insole for continuous gait analysis in rehabilitation

A shoe insole that measures gait (pattern of movement) using the energy of each footstep: a triboelectric sensor, a custom signal conditioning electrometer front-end, an ESP32 acquisition board and an XGBoost classifier, built end to end.

TENG · Analog front-end · OrCAD PCB · ESP32 · AD795 · TLV2772 · Python · FFT · XGBoost · scikit-learn

93.58%	0.94	12 subjects	≈800 mV
Accuracy across 4 activities on 122,652 held-out windows	Macro F1-score; per-class F1 between 0.93 and 0.95	84 recordings: standing, walking, running, jumping	Open-circuit voltage; 10.58 µW peak harvested power

THE PROBLEM

Clinical gait analysis happens in a lab and captures a short snapshot. About 2.4 billion people live with a condition that could benefit from rehabilitation, yet how a patient actually walks at home between appointments goes unmeasured.

MY APPROACH

Use a triboelectric nanogenerator (TENG) as a *self-powered* pressure transducer in the heel, design electronics that can read its GΩ-impedance output without destroying it and classify activity from frequency-domain features.

THE RESULT

A validated proof of concept, from sensor to classification, that reached 93.58% accuracy and showed the TENG can serve as both sensor and energy source. A journal paper is under review at *IEEE Sensors Journal*.

SYSTEM ARCHITECTURE

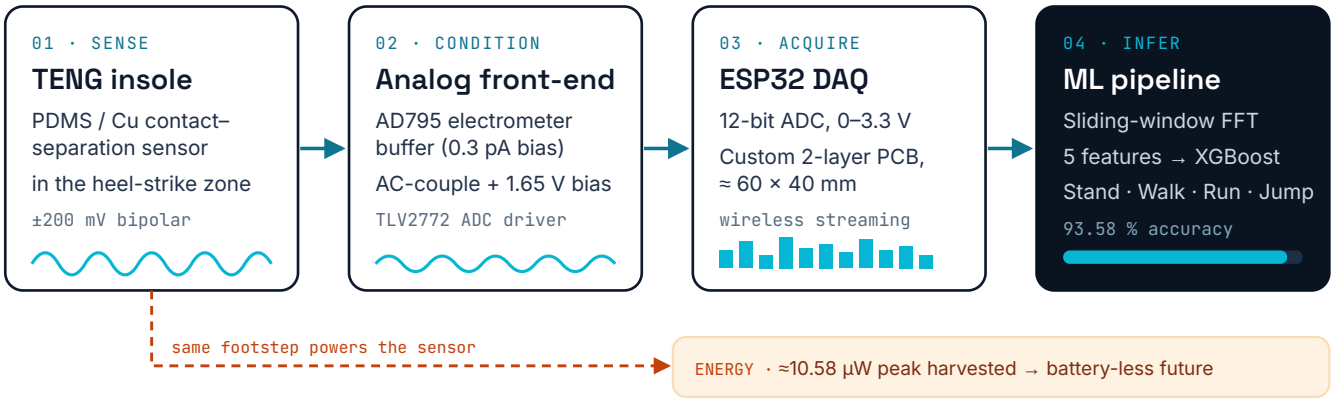


FIG 1.1 The signal path from heel strike to activity label. The dashed path marks the second role of the TENG, as an energy harvester, which the thesis characterised as groundwork for a fully battery-less version.

R My role

Sole author of the thesis. I specified the system, designed and laid out the signal-conditioning PCB, wrote the acquisition firmware, ran the 12-participant study and built the ML pipeline. The PDMS dielectric material was developed by my supervisor's group.

S Supervision

Prof. Dr.-Ing. Peter Woias (first examiner), Dr.-Ing. Laura Maria Comella (second examiner) and adviser Fiza Asif, Laboratory for the Design of Microsystems, IMTEK, with livMatS Freiburg.

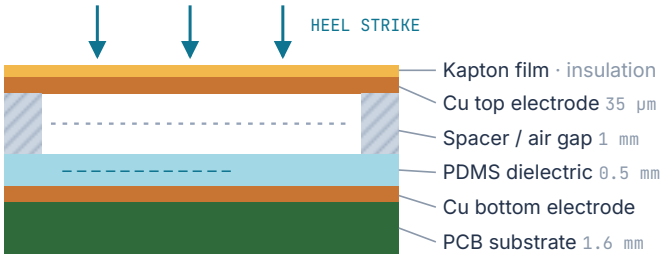
01.1 Hardware: sensor and front-end

SENSOR · AFE · PCB

A self-powered transducer

The sensor is a contact-separation TENG. When the heel strikes, copper and PDMS touch and exchange charge; as they separate, the changing gap drives current through the electrodes. No battery is needed to *sense*, because each footstep generates the signal.

- Active area 2 × 2 cm, sized to the heel's peak ground-reaction-force zone
- Layer count optimised experimentally, from 3 to 6 layers, for amplitude
- Encapsulated between EVA foam insole layers against moisture and abrasion



Cross-section, not to scale · PDMS: Sylgard 184, 10:1

FIG 1.2 TENG layer stack. Heel strike closes the 1 mm gap; separation on toe-off drives the charge back.

Solving the impedance problem

A TENG is a very high-impedance source (above 10 MΩ, effectively GΩ). Connected straight to an ADC, the input would load it and destroy the signal. I designed a three-stage conditioning chain that preserves the waveform and maps the bipolar ±200 mV output into the ESP32's 0–3.3 V ADC range.

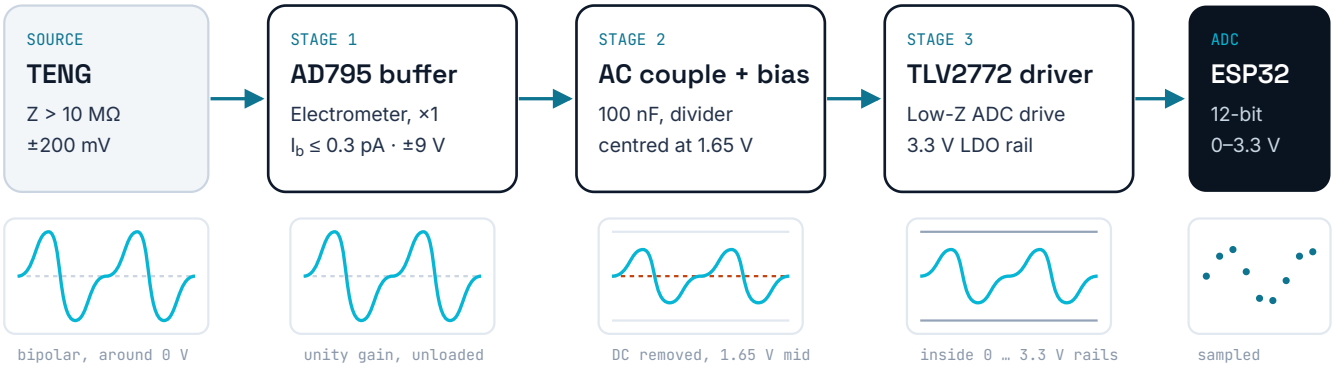


FIG 1.3 Three-stage conditioning chain with the idealised waveform at each node, based on the simulated node voltages in the thesis.

Key design decisions

- **Electrometer first.** At ≤0.3 pA bias current, the error across a 10 MΩ source stays below 50 µV, under 0.01% of the signal.
- **Mid-rail biasing.** AC coupling plus a 1.65 V reference centres the bipolar waveform so neither half clips at the ADC.
- **Band-limited to gait.** Gait lives in roughly 0.5–5 Hz, so the filtering rejects high-frequency noise without distorting strides.
- **Split power domains.** ±9 V for the precision stage and a 3.3 V LDO for the ADC side, with a single-point analog/digital ground.
- **Modular connectors** (TENG in, battery, ADC out) so the sensor, analog and digital blocks can be swapped and tested independently.

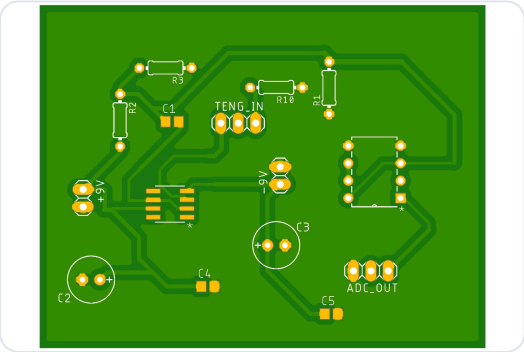


FIG 1.4 Two-layer signal-conditioning PCB (≈60 × 40 mm): TENG_IN, AD795 stage, ±9 V inputs and ADC_OUT header.

01.2 The built prototype

FABRICATION · INTEGRATION · STUDY

The design left the bench as a working wearable: a hand-fabricated TENG sensor, integrated into a standard shoe insole and worn by 12 participants for 84 recorded trials.

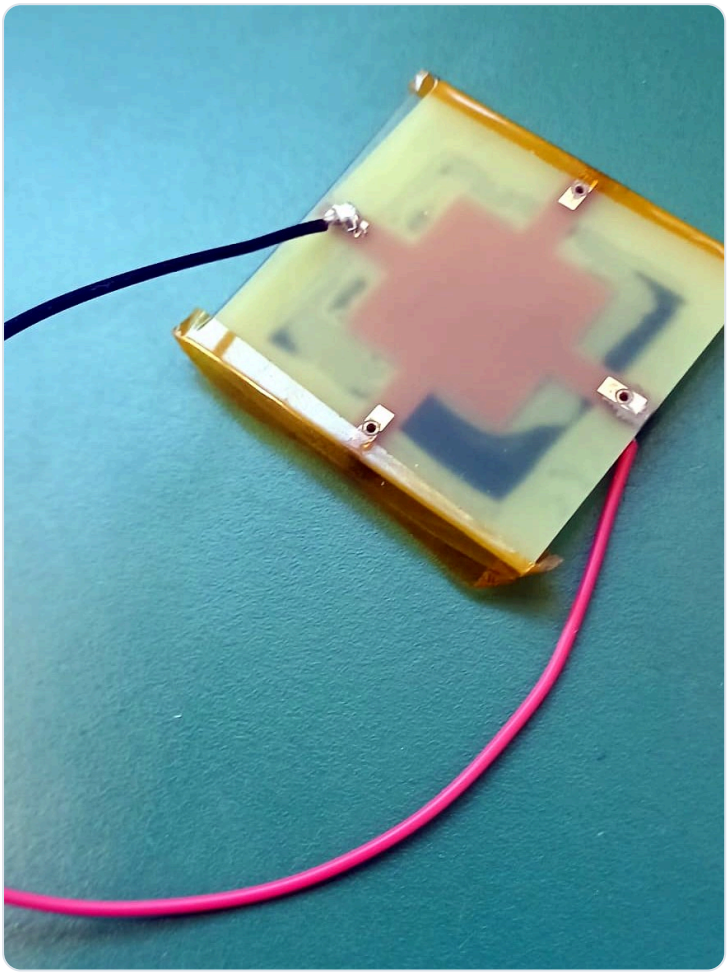


FIG 1.5 Fabricated TENG sensor: Kapton-covered copper top electrode over the PDMS dielectric and PCB-mounted bottom electrode, with leads to the conditioning board.



FIG 1.6 The sensor integrated into the right insole's heel region; the leads run to the conditioning circuit at the ankle.

FROM BENCH TO STUDY

1 Fabricate
PDMS (Sylgard 184, 10:1) spin-coated and cured on the copper bottom electrode; a Kapton-laminated copper top electrode over a 1 mm spacer.

2 Optimise
Layer count increased from 3 to 6 to raise the signal, reaching ≈800 mV open-circuit and 10.58 µW peak power.

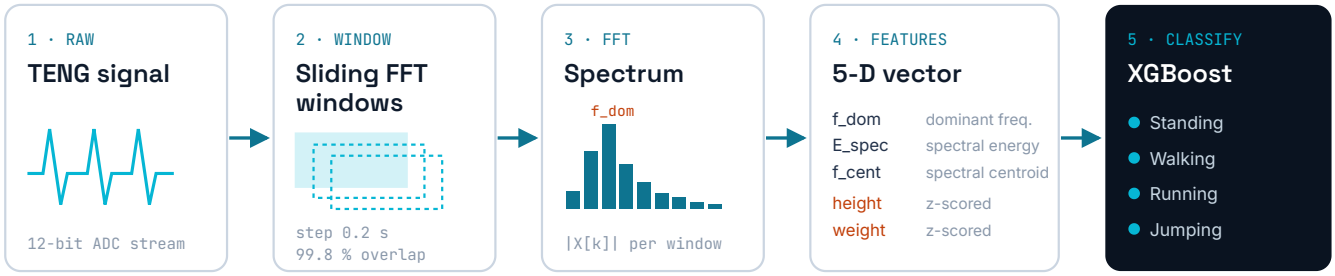
3 Integrate
Sealed between EVA foam insole layers at the heel; silicone-insulated AWG 30 leads taped along the shoe to an ankle enclosure.

4 Test
12 participants × 7 trials (stand, walk ×2, run ×2, jump ×2) = 84 recordings, feeding the classifier on the next page.

01.3 From waveform to activity label

FEATURES · MODEL · EVALUATION

TENG waveforms are quasi-periodic and sensitive to contact mechanics, so time-domain peak counting is fragile. I moved the problem into the frequency domain, where each activity has a distinct spectral signature, and chose a gradient-boosted tree model that trains in minutes, explains its decisions and is small enough for edge deployment.



`multi:softprob · 1000 estimators · max depth 6 · η = 0.3 · λ = 1.0 · stratified 75/25 split`
`366,956 training windows · 122,652 test windows · from 12 participants`

FIG 1.7 Classification pipeline. Probabilistic outputs (softprob) give a confidence score per window, useful for flagging uncertain segments in deployment.

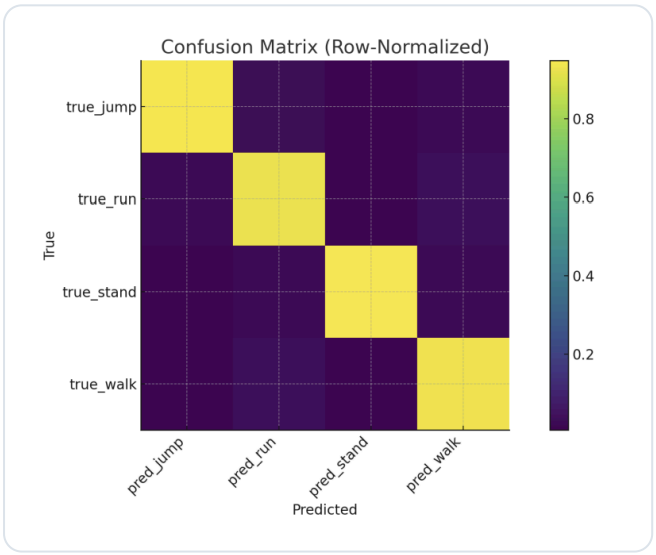


FIG 1.8 Row-normalised confusion matrix. Standing reaches 95% recall; the main confusion (4%) is walking vs running, which sit on a biomechanical continuum.

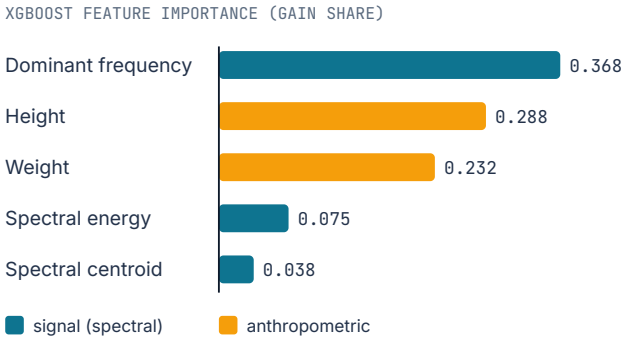


FIG 1.9 Dominant frequency tracks cadence (1–2 Hz walking, 2–4 Hz running). Height and weight together carry about 52%, because body mass shapes the force profile the TENG sees.

ACTIVITY	PRECISION	RECALL	F1	SUPPORT
Jumping	0.95	0.94	0.95	32,112
Running	0.92	0.93	0.93	36,216
Standing	0.93	0.95	0.94	18,108
Walking	0.94	0.93	0.93	36,216
Macro avg.	**0.94**	**0.94**	**0.94**	**122,652**

WHY GRADIENT-BOOSTED TREES RATHER THAN AN LSTM

Training cost

Trains in minutes rather than hours, so feature ideas could be tested and compared quickly.

Interpretability

Explicit feature importance shows *why* a window was classified, which matters in clinical settings.

Deployability

Five input features and a tree ensemble map well onto a microcontroller for on-device inference.

01.4 Critical review and next steps

What I would tighten

- **Subject-level validation.** The 75/25 split was made at window level, so it can be optimistic. Leave-one-subject-out cross-validation is the honest next benchmark.
- **Identity vs activity.** Height and weight carry about half the importance. I want to confirm the model generalises to unseen bodies and is not partly recognising individuals.
- **Dataset scale.** 12 healthy adults in a lab. The clinical question needs patients, more diverse users and in-the-wild data.

Where it goes next

- **Full energy autonomy.** Add on-board harvesting and power management so the insole runs without a battery.
- **Sensor fusion.** Pair the TENG with a low-power IMU for richer gait-phase and quality-of-movement metrics.
- **On-device inference.** The compact tree model makes ESP32 classification feasible, closing the loop to real-time feedback for patients and therapists.

Output: *"Intelligent Biomedical Wearable for Gait Analysis in Rehabilitation"*, submitted to IEEE Sensors Journal (under review). Code and documentation: `github.com/ajeethv27/teng-gait-wearable`

PROJECT 02 STUDENT RESEARCH ASSISTANT · FRAUNHOFER EMI, FREIBURG · FEB – JUL 2024

Autonomous mobile rover with LiDAR, IMU and camera fusion

A ROS-based ground rover that navigates on its own by combining three complementary sensors, with the low-level control written in C++ on Linux.

ROS | Sensor fusion | C++ | Linux | LiDAR | IMU | Camera | Filtering

95%	+30%	−40%
Autonomous navigation success	Sensor performance through advanced filtering	System error rate after rewriting low-level control in C++

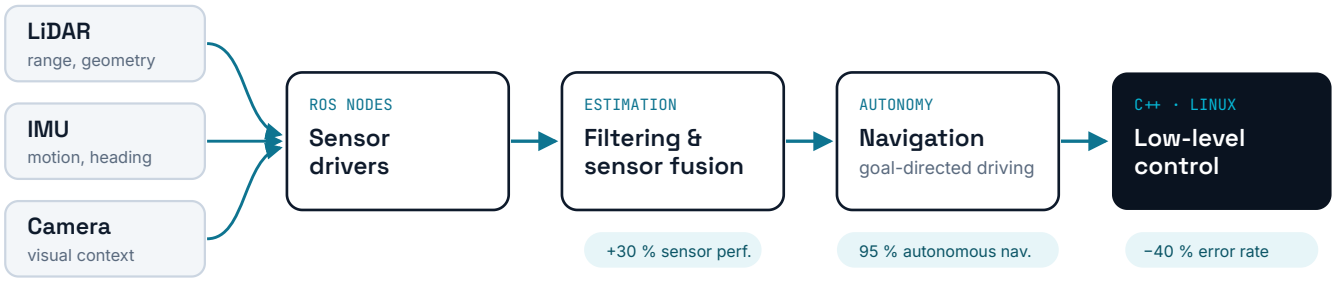


FIG 2.1 Functional architecture of the rover software stack.

Internal Fraunhofer research project: source code, photos and data remain with the institute. I am happy to discuss the architecture and my contributions in detail in an interview.

PROJECT 03 START-UP PROJECT · HAHN-SCHICKARD / UNI FREIBURG · AUG 2023 – JAN 2024

Smart waste monitoring with TinyML

A low-power motion-tracking node that recognises bin events on the device itself and alerts the facility team in real time, so collection responds to real events instead of fixed rounds.

TinyML | Arduino Nano 33 BLE | Accelerometer | Signal filtering | MQTT | Low power

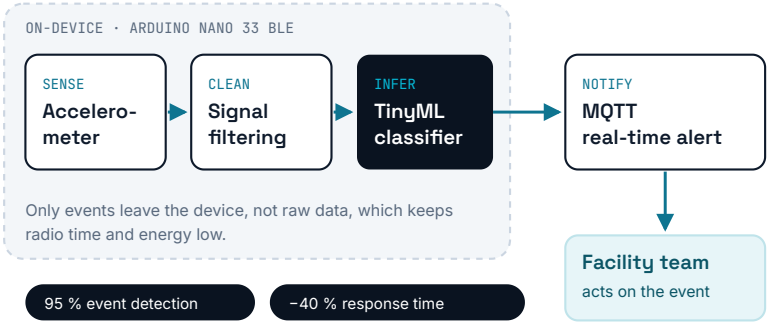


FIG 3.1 Edge-first architecture: sense, filter and classify on the microcontroller, then publish only the event.

Why edge inference

Streaming raw accelerometer data from every bin would drain batteries and flood the network. Running a compact TinyML model on the Nano 33 BLE means only detected events need to be transmitted.

- Collected and labelled motion data, then trained a model small enough to run on the microcontroller
- Filtered the accelerometer signal before inference to make event detection robust
- MQTT alerts cut facility response time by 40%

PROJECT 04 FELLOWSHIP & RESEARCH · FORGE INNOVATION & VENTURES · JUL – DEC 2019

Wireless IoT energy management for smart factories

An ESP32 wireless sensor network with PCB-mounted current sensors that shows where a facility's energy goes and flags anomalies for predictive maintenance.

ESP32 | Current sensing PCB | 2.4 GHz WSN | Machine learning | Predictive maintenance

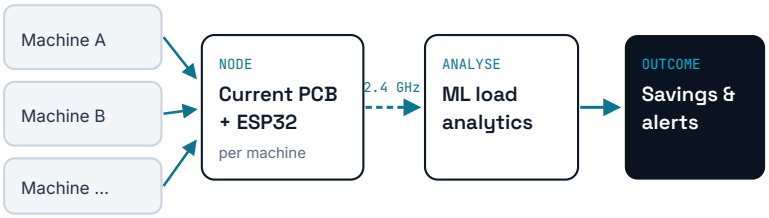


FIG 4.1 Metering nodes on each load report over a 2.4 GHz network to an analytics layer.

30–50 % Energy savings identified in industrial facilities (fellowship)

20 % Energy savings in the research prototype

Published as *"Energy Management System for Smart Factories"*, IRJET, Vol. 7, Issue 9, 2020, pp. 3390–3393.

PROJECT 05 B.E. THESIS, TEAM OF THREE · KUMARAGURU COLLEGE OF TECHNOLOGY · JAN – JUN 2021

IoT-based sewage monitoring and alert system

A sensor node that detects blockages, overflow and hazardous gas in drains and pushes live readings, location and alerts to a municipal worker's phone, so crews arrive before an overflow becomes a health hazard.

Arduino Uno | ESP8266 NodeMCU | MQ-2 gas | Flow · level · leakage | GPS | Blynk | PCB design | C++ (Arduino)

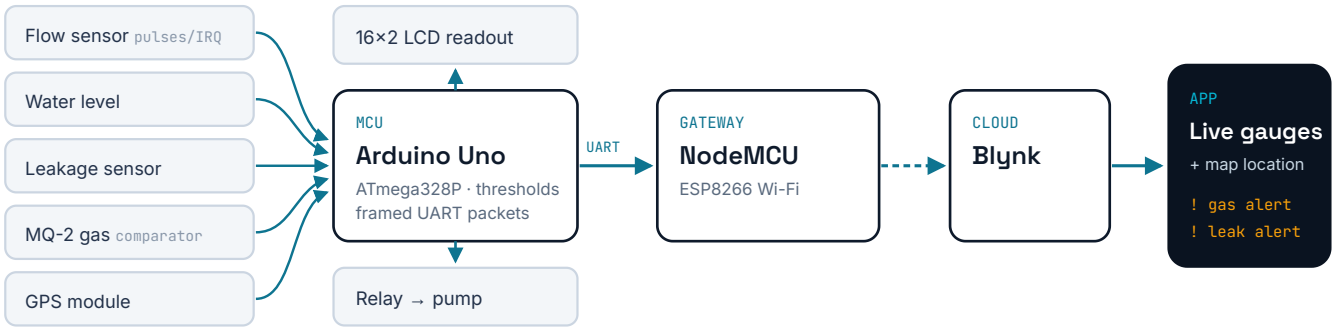


FIG 5.1 Node architecture. The Uno samples and thresholds all sensors, drives the local display and pump relay, and sends a framed packet to the ESP8266, which publishes to Blynk and raises push notifications.

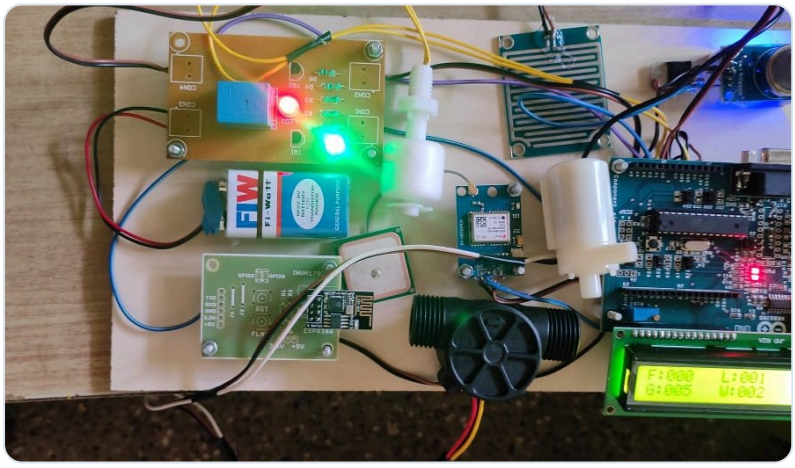


FIG 5.2 Working prototype: Arduino Uno with LCD, flow sensor, gas, level and leakage sensors, GPS, NodeMCU and custom relay/power boards.

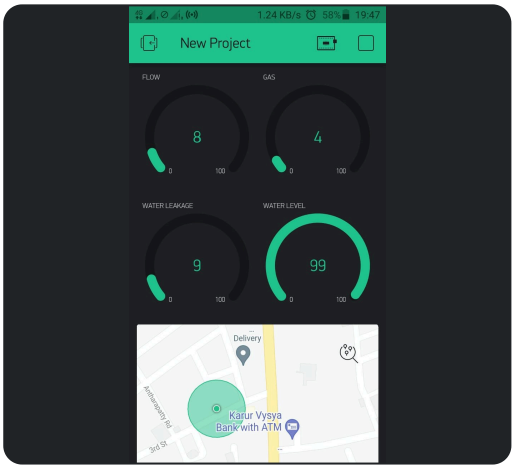


FIG 5.3 Blynk dashboard with live gauges and node location.

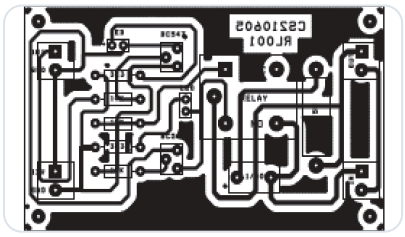


FIG 5.4 Relay driver PCB (BC547 pair).

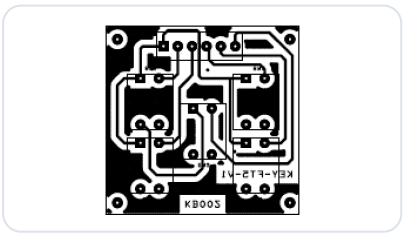


FIG 5.5 Keypad interface PCB.

What we built

- Interrupt-driven flow measurement (L/min and total volume)
- Threshold alerts for gas, leakage and low level, plus automatic pump control
- A compact serial protocol from the Uno to the ESP8266
- Single-sided PCBs for the power, relay and keypad stages

Code ↗ github.com/ajeethv27/iot-sewage-monitoring

06 Professional experience

2019 – 2025

Software Engineer · Cognizant

BENGALURU · MAR 2021 – AUG 2022

TEST AUTOMATION · BANKING

Built a **Selenium WebDriver** automation framework in **Java** for Temenos core-banking software, replacing repetitive manual regression runs with maintainable automated suites.

−45%
Manual testing cycle time

DATA ANALYTICS · HEALTHCARE

Ran biostatistical analysis of clinical data in **Python** (NumPy, pandas) and designed **Power BI** dashboards that made healthcare KPIs visible to stakeholders.

`Java` `Selenium WebDriver` `Temenos` `Python` `NumPy` `pandas` `Power BI`

> Client work under NDA; no source code can be shared.

Master's Thesis Student · livMatS

FREIBURG · NOV 2024 – NOV 2025

- Precision signal-conditioning PCB in OrCAD with electrometer buffer and ESP32
- XGBoost pipeline with FFT features: 94% accuracy for clinical gait assessment
- TENG insole (PDMS/Cu) demonstrating battery-free sensing potential

→ PROJECT 01, P. 03

Student Research Assistant · Fraunhofer EMI

FREIBURG · FEB – JUL 2024

- ROS rover with LiDAR, IMU and camera fusion: 95% autonomous navigation
- Advanced filtering improved sensor performance by 30%
- Low-level control in C++ on Linux cut system error rates by 40%

→ PROJECT 02, P. 07

Start-up Project · Hahn-Schickard

FREIBURG · AUG 2023 – JAN 2024

- TinyML motion tracking on Arduino Nano 33 BLE: 95% event detection
- MQTT alerts reduced facility response time by 40%

→ PROJECT 03, P. 08

Graduate Innovation Engineer · Forge

COIMBATORE · JUL – DEC 2019

- IoT energy monitoring with PCB-mounted current sensors and ML
- 30–50% energy savings in industrial facilities

→ PROJECT 04, P. 08

07 Technical skills

GROUPED BY DISCIPLINE

Programming — SOFTWARE
C / C++ · Embedded C · Python · MATLAB · Java

Embedded platforms — MCU / OS
ARM Cortex-M · STM32 · ESP32 · ESP8266 · Arduino · FreeRTOS · Linux · ROS

AI / ML — EDGE AI
XGBoost · Gradient-boosted trees · scikit-learn · TensorFlow · PyTorch · TinyML · FFT features

Hardware — ANALOG / PCB
Analog signal conditioning · Electrometer front-ends · Power electronics · OrCAD · Altium

Sensors — TRANSDUCERS
IMUs · TENG energy harvesters · LiDAR · Current sensing · Gas / flow / level

Communication — PROTOCOLS
CAN · SPI · UART · I²C · Bluetooth / BLE · Wi-Fi · MQTT · LoRaWAN

Medical devices — MEDTECH
Patient-safe low-voltage design · EMC for biomedical sensors · PoC prototyping

Tooling — DEV
Git · Docker · VS Code · Selenium · Power BI

Languages — SPOKEN
English C1 · German B1 · Hindi B2 · Tamil native

Evidence matrix

Instead of self-rated percentages: where each skill was actually used. ● core to the project ○ supporting role.

Skill area	01 TENG wearable	02 ROS rover	03 TinyML waste	04 Energy WSN	05 Sewage IoT	Cognizant
Analog signal conditioning	●			○	○	
PCB design	●			●	●	
Embedded C/C++ firmware	●	●	●	●	●	
Linux · ROS		●				
Sensor fusion & filtering	○	●	●			
Signal processing (FFT)	●					
Machine learning	●		●	●		○
Wireless & IoT cloud	○		●	●	●	
Energy harvesting / low power	●		●	○		
Test automation & QA						●
Data analysis & dashboards	●			○	○	●

08 Publications

2025
UNDER REVIEW

Intelligent Biomedical Wearable for Gait Analysis in Rehabilitation

IEEE Sensors Journal, submitted manuscript based on the M.Sc. thesis

Presents a TENG-based smart insole that captures gait through triboelectric sensing, a custom signal-conditioning circuit for low-power 3.3 V ADC systems, and an XGBoost pipeline that classifies standing, walking, running and jumping with 94% accuracy, while showing the TENG's potential as an energy source for battery-less health monitoring.

2020
PUBLISHED

Energy Management System for Smart Factories

International Research Journal of Engineering and Technology (IRJET), Vol. 7, Issue 9, pp. 3390–3393

Describes an ESP32-based wireless sensor network with PCB-mounted current sensing for monitoring industrial loads, supporting predictive maintenance and measurable energy savings.

08.1 Education

OCT 2022 – MAR 2026

M.Sc. Embedded Systems Engineering

Albert-Ludwigs-Universität Freiburg, Germany

Thesis: Intelligent and Energy-Autonomous Biomedical Wearable Platform for Continuous Gait Analysis in Rehabilitation (IMTEK, Laboratory for the Design of Microsystems)

Circuits & Systems · Biomedical Engineering · Renewable & Energy Harvesting · Artificial Intelligence

AUG 2017 – JUN 2021

B.E. Electrical & Electronics Engineering

Kumaraguru College of Technology (Anna University), India

Thesis: IoT-Based Sewage Monitoring System

Embedded Systems · AIoT · Industrial Design & Product Development · Applied Design Thinking · Signal Processing · Microcontrollers

08.2 Languages

English
FLUENT · C1

German
INTERMEDIATE · B1

Hindi
INTERMEDIATE · B2

Tamil
NATIVE

Bars show CEFR level (A1 → C2).

09 How I work

1 Start from the physics

Before choosing parts, I model the source. The TENG's electrostatic model is what made the electrometer front-end an obvious choice.

EVIDENCE → PROJECT 01

2 Design around the signal

Impedance, bandwidth and headroom decide the architecture. Biasing to mid-rail and band-limiting to the gait band kept the ADC data clean.

EVIDENCE → PROJECT 01

3 Compute at the edge

Send decisions, not raw data. On-device TinyML and compact tree models save radio time, energy and bandwidth.

EVIDENCE → PROJECT 03, 01

4 Prefer models I can explain

For tabular sensor features, gradient-boosted trees train in minutes and show which feature drives each decision, which matters in clinical settings.

EVIDENCE → PROJECT 01

5 Automate the boring parts

Repeatable tests beat heroic manual runs. At Cognizant, automated regression suites cut manual testing by 45%.

EVIDENCE → COGNIZANT

6 Be honest about validation

I document limits alongside results, such as window-level vs subject-level splits, so the next iteration fixes the right thing.

EVIDENCE → P. 07

Impact at a glance

93.6%	**95%**	**95%**	**−45%**
Gait classification, self-powered TENG insole	Autonomous navigation, ROS rover (Fraunhofer EMI)	Event detection, TinyML waste node	Manual testing cycles (Cognizant)

> I am looking for roles where embedded hardware, signal processing and machine learning meet, ideally in medical technology, wearables or intelligent IoT devices and where a prototype has to become a reliable product while improving the quality of life of all beings.

Let's talk

Open to embedded systems roles in Germany. Source code, schematics and demos for the public projects are on GitHub; I'm happy to walk through the rest in person.

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